ML RESEARCH ARTIFACT · FLYRANK INTERNSHIP

Content Opportunity Scoring for Search Performance Review

Using historical search signals to prioritize content pages for human review

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PRECISION@50 — TOP 50 PAGES SELECTED, SHARE MATCHING THE DECLINING-PERFORMANCE OUTCOME

BASELINE	0.240
RANDOM FOREST	0.740

Client-grouped validation · **+0.500** absolute lift over baseline

This project investigates whether historical search-performance and query signals can help prioritize content pages for human review and possible refresh. Using the FlyRank ML Internship dataset, I constructed historical features describing impressions, query visibility, query concentration, and ranking behavior, then compared a Random Forest model with a transparent baseline. The model was evaluated using client-grouped validation to reduce optimistic estimates caused by having pages from the same client in both training and testing data. The results provide directional evidence that these signals can support content-review prioritization, although the constructed declining label is only a proxy and performance may vary under stricter validation. The final output is a ranked review queue with reason codes and suggested actions, intended to support human content decisions rather than automate publishing or claim causal effects.

§01 Introduction / Problem Statement

Content teams often have more pages to review than they can investigate manually. A useful system should therefore help identify which pages deserve attention first.

> **Can historical search-performance and query signals provide useful decision support for prioritizing content pages for review?**

The goal is not to automatically decide which pages should be changed. Instead, the goal is to produce a ranked queue that helps a human reviewer allocate limited time.

§02 Data

This project uses the FlyRank ML Internship warehouse release. The analysis uses historical search-performance and query-level signals to construct content-level features over a 90-day window.

The data is pseudonymized and the public artifact does not expose client names, URLs, private search queries, or other identifying information.

EXCLUDED FIELDS

Label-derived fields such as `trend_direction`, `trend_pct`, and `is_declining_label` were excluded from model features because they contain or directly derive information about the outcome and could introduce leakage.

§03 Methodology

TASK

The task is scoring and ranking content pages for review using a binary declining-performance proxy for evaluation.

FEATURES

- `imp_mid30`
- `visible_queries`
- `rare_share`
- `anon_share`
- `top_query_share`
- `pos_volatility_last30`

MODEL

A Random Forest classifier was used because it can capture nonlinear relationships between the historical search signals and the constructed declining-performance outcome.

VALIDATION

The final evaluation uses client-grouped validation. Pages belonging to the same client are kept entirely within either the training or testing partition.

§04 Results

BASELINE · PRECISION@50

0.240

reference rate

RANDOM FOREST · PRECISION@50

0.740

+0.500 vs. baseline

The baseline achieved a Precision@50 of 0.240, meaning that 24% of the top 50 pages selected by the baseline matched the defined declining-performance outcome.

The Random Forest achieved a Precision@50 of 0.740, meaning that 74% of the top 50 pages selected by the model matched the defined outcome.

IMPORTANT

Precision@50 is not the same as overall accuracy. The 0.740 result means that 74% of the model's top 50 ranked pages matched the defined outcome. It does not mean that the model is 74% accurate across the entire dataset.

MODEL	PRECISION@50
Baseline	0.240
Random Forest	0.740

§05 Limitations & Honest Framing

- The declining-performance label is a constructed proxy.
- The analysis is observational and does not establish causation.
- The model does not predict Google's ranking algorithm.
- Performance may vary across clients, topics and future periods.
- A high model score does not mean a page should automatically be changed.

The results should therefore be interpreted as observed, measured, directional decision support rather than causal proof or guaranteed prediction.

§06 Ranked Recommendations

1. **Declining impressions**
   Review whether the content remains accurate, useful and aligned with current search intent. Consider a refresh only after human review.

2. **Position volatility**
   Review ranking changes and SERP alignment.

3. **Query concentration**
   Review whether the page depends too heavily on a small number of queries.

4 **Low query visibility**

Review targeting, discoverability and topic alignment.

! **Manual review**

Pages receiving high model priority without a clear reason should be manually inspected before any action.

§07 Human Review & No-Go Cases

The model is a decision-support system. Human review is required before changing content.

> The system should not automatically publish, delete, rewrite, or modify pages. It should also not claim that a model score proves why a page declined.

§08 Reproducibility

The complete analysis is available in the project repository. The notebooks document the progression from research question and data contract through baseline modeling, validation and the final action playbook.

`View the complete GitHub repository →`

§09 Acknowledgments & Data Credit

Built on the FlyRank ML Internship dataset.

Data source: FlyRank

I thank the FlyRank ML Internship team for providing the dataset, learning framework and project structure used for this work.